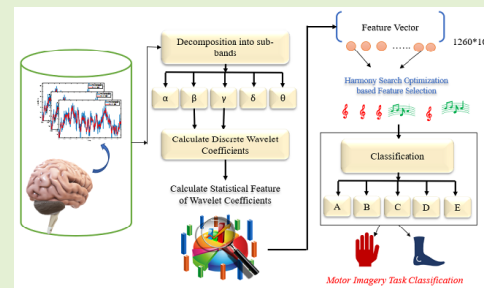


Statistical Wavelets With Harmony Search-Based Optimal Feature Selection of EEG Signals for Motor Imagery Classification

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Abstract—Brain Computer Interface (BCI) does not only help people of physical disability but also being used popularly in several other applications. Motor Imagery (MI) classification is one of the major contributions in BCI which works on segment of EEG signal within particular frequency band. Herein, feature selection plays important role in obtaining good classification results. In this article, Beta and Gamma frequency are considered with Statistical DWT (SDWT) based features for classification of EEG signals (MI classification) for patient monitoring, assistance healthcare services and daily living activities. Harmony search algorithm of feature selection is used to obtain the optimal feature set for classification of MI. The results show that frequency centric SDWT achieves average accuracy of 92.49% for weighted KNN (K-Nearest Neighbour) method. Comparison of accuracies before and after feature selection portrays that feature selection with harmony search improves the performance of proposed MI classification.

Index Terms—Brain computer interface, EEG, motor imagery, wavelet transform, optimal feature selection, harmony search optimization.



I. INTRODUCTION

THE electroencephalograph (EEG) is the most prominent, non-invasive and less costly signal acquisition technique which is used in brain computer interface (BCI) for communication between brain and machine. Communication for imagined motor movements such as hand or feet movement from brain to system is termed as motor imagery task for BCI which is useful in assistive robot technologies controlled by

brain and used in several applications [1]–[3]. It is interesting to note that in motor imagery (MI) task classification, no external stimulus is required to perform imagination task, which is the most important and sensitive part of other BCI based applications such as mental arithmetic and speller designs. In motor imagery task classification for BCI, EEG signals are mostly used due to their non-invasive nature of acquisition which results in no cortical damage in brain than other invasive technique but suffers from high artifacts. These artifacts result in reduced accuracy for BCI performance. The EEG signals are non-stationary in nature and become a challenging task to work with these signals and to mitigate this, the signals are segmented into small time frames in which the signal is assumed to be stationary. Various studies have been reported for MI task classification [4], [5] etc. using time domain, frequency domain and common spatial pattern as features for motor imagery classification task. In spite of numerous techniques on motor imagery classification reported in current literature, there remains a potential scope in the area best suitable technique that does not only classify the MI tasks but also reduces the complexity of the model and the feature space. It is important to extract the relevant features simultaneously

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to reduce the irrelevancy of features along with feature space. To work on this, we have studied different features and feature selection techniques which are applied in various applications. Based on the study, a method is proposed to effectively classify the MI tasks. The details are provided further in the paper.

The paper is organized as follows: Section 2 presents related work; In section 3, Motivation and Problem Formulation are discussed; Section 4 describes the datasets used along with proposed method of sub-band based statistical wavelet for feature extraction and harmony search based feature selection method. Section 5 discusses the results of proposed method with analysis of the results and comparison with other existing method. Finally, the section 6 concludes the paper.

II. RELATED WORK

A. Related Work for MI Feature Extraction

Common spatial pattern (CSP) is widely used technique for Motor imagery task classification which works on segment of EEG signal with particular frequency band. The concept of CSP is referred as optimization problem for maximizing the variance of positive class and simultaneously minimizing the variance of negative class [6]. The CSP features focus on spatial property of the signal not on the spectral one and therefore choice of frequency bands and specific frequencies becomes important issue in common spatial patterns [7], [8]. The CSP also suffers from problem of overfitting because it is based on L-2 norm method [6], [9] that can be overcome by Regularized CSP and L-1 norm based CSP's [10], [11]. These CSP methods are computationally intensive.

Considering the above CSP aspects and limitations of various methods, frequency or time frequency characteristics of motor imagery EEG signal are matter of concerns. While performing MI task, event related synchronization and desynchronization take place in sensory motor rhythms mainly in $[\mu, \beta]$ and γ frequency band respectively [5], [12]. Power Spectral Density (PSD) is also highly explored feature in MI task categorization.

Further, we investigated literature for support on frequency based features or their extension in addition to CSP based methods. Park *et al.* [13] proposed frequency based multi-variate empirical mode decomposition to localize time frequency representation. Kevric and Subasi [14] demonstrated the comparative analysis of decomposition based methods and found that higher order statistics features are more prominent over other features extracted from DWT, WPD with sub-bands. It also provides comparative analysis of DWT and WPD based method with CSP based approach and found that wavelet based approach more consistent in terms of performance. Mirvaziri and Mobarakeh [15] adopted DWT feature extraction technique and attempted to improve the classifier with PSO based algorithm. Luo *et al.* [16] investigated the effect of ERS/ERD and spatio-temporal discrepancy feature is suggested for improvement in signal to noise ratio (SNR). Welch power spectral density, Burg's Auto-regressive feature, and Wavelet packet decomposition are extracted and classified with Support Vector Machine.

Some recent techniques on MI task classification include multiplex weighted visibility graph [17]; energy principal component analysis based feature extraction and deep belief network [18]; and improved channel selection for CSP MI task classification [19]. However, these techniques have shortcomings that either it required subject wise calibration or uses deep learning technique which improves the overall results on the expense of complexity of the model. The small training samples generates the need of effective and suitable data augmentation technique.

B. Related Work for Feature Selection and Optimization Techniques

Feature selection is one of the important aspect that is useful to reduce the feature space and complexity as well as to get the relevant features. Based on taxonomy of feature selection, evaluation strategy of feature selection can be categorized into 5 major categories i.e. Filter method, Wrappers method, Embedded method, Hybrid method, Ensemble method [20]. Filter method is dependent on selection criteria such as information, distance, dependency etc however, it is independent of learning algorithm. Minimum redundancy maximum relevancy (mRMR) [21], entropy, information gain [22], Relief-F [23], Chi-Square [24] are some of the techniques of filter method. This technique is used in the channel selection and feature selection problems for different EEG based applications such as emotion recognition [25], stress detection [26] and many more. Wrappers method uses the dependency among the features and it is classifier specific. Different combinations of features are tested on the model and provide the best combination based on the forward or backward selection [20]. Optimization method can also be used with wrappers method to provide the optimal subset of features. Feature selection with wrapper method is used in various literature for EEG tasks categorization [27], [28].

Hybrid and Ensemble methods are also popular in few applications. When optimization based methods are considered for MI tasks, to improve the diversity of ensemble classifier, sub-band based CSP with Fuzzy integration of PSO is suggested which introduced information of uncertainty, incompleteness and imprecision of EEG, generally skipped during feature extraction, via fuzzy integration method. Fuzzy method PSO is used for improved confidence [29]. Another study of optimization based MI task classification uses Differential Evolution method for feature selection. Wrappers method of feature selection best suits to check the performance of optimization algorithm [30] Harmony search optimization is one of the popular optimization technique nowadays due to its limited mathematical requirement and successful due to its insensitiveness to initial value. The method proved its efficiency in various machine learning problems and shows various advantages over existing traditional optimization methods [31] hence, this technique is further utilized in the current work.

III. MOTIVATION AND PROBLEM FORMULATION

A comprehensive study on MI task classification using different methods has been discussed so far. The study suggests the need of a model that can effectively classify MI task for

different applications such as patient monitoring, home and healthcare assistance. Advantages of frequency based features over CSP features enhances the interest towards frequency and time - frequency based feature extraction technique. Popularity of optimization techniques due to successful implications in different fields motivate to use the technique for MI task classification for performance improvement.

Keeping this in notion, the paper proposed sub-band selection based statistical wavelet features for MI task classification. Sub-bands referred to β and γ band represent the characteristics of MI task [32]. We aim to: (1) Improve classification accuracy for MI task and (2) Analysis of optimization technique based feature selection for MI tasks. Wrappers method of feature selection with harmony search optimization has been employed for task characterization. Moreover, the features obtained from selection technique are best suitable, further utilized in different classifiers to improve performance. Finally, comparison between proposed method and different existing methods is done based on accuracy.

IV. MATERIALS AND METHOD

This section discusses dataset used and proposed methodology for MI task classification which includes description of dataset, methods for feature extraction, feature selection and classification.

A. Dataset

The current study is based on open source datasets available for MI task classification. The dataset is taken from BNCI Horizon 2020 website which dataset consists of 10 subjects. Cue guided grazed paradigm was used for recording. The task performed in a single session contains 8 runs (5-Training and 3-Validation), each run having 20 trials. The subjects were asked to perform kinesthetic MI of right hand and feet, sustained for 5 sec. The recordings are taken from Ag/AgCl electrodes placed on C3, C4, Cz positions and neighbouring positions on the central lobe. Detail description of dataset is discussed for further understanding in [33], [34].

B. Proposed Method

The workflow of proposed method is shown in Fig. 1 which highlights the overview of MI task classification. The methodology consists of three steps: initial step is data composition in which data is prepared as per the need of experimental design, second step include details of feature extraction method and in final step, feature selection classification and performance evaluation are performed. Each block of proposed method is discussed in detail in further subsections.

1) Data Decomposition: The experimental data has been imported into the workspace for further processing.

Data taken from electrodes is unprocessed and contain various artifacts; one such artifact is human function artifact which can be reduced during preprocessing stage from the original signal. In this stage, undesired frequencies and information are removed. To preprocess the data, 5th order Butterworth bandpass filter is employed which is having

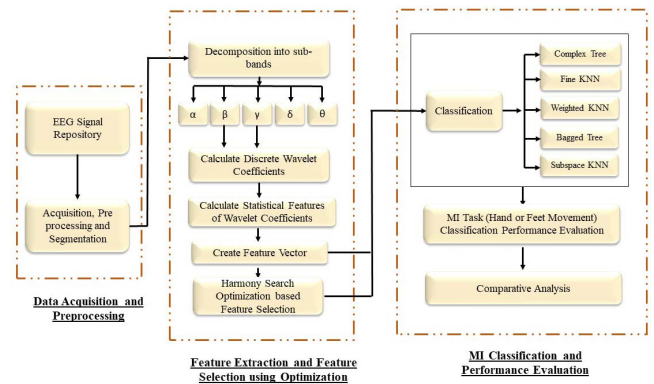


Fig. 1. Block diagram of proposed methodology.

passband frequency of 0.5Hz and Stopband frequency 100Hz. Maximally flat response and uniform passband property of this filter makes it more suitable at the cost of wide transition band. The Butterworth filter is mathematically expressed as (1).

$$H(x, y) = 1 - \frac{1}{1 + \left[\frac{D(x, y)W}{D(x, y)^2 - D_0^2} \right]^{2n}} \quad (1)$$

where, $D(x, y)$ is the distance from center frequency of window used from points x, y ; W is the bandwidth of filtering range; and n is the order of the filter. Choice of order of the filter depends upon the tolerance of transition band as well as number of datapoints available per sampling frequency. After preprocessing operation, the data is segmented according to the values provided in the description of dataset for each task for both the dataset.

2) Feature Extraction: Signal obtained from segmentation procedure has high dimension and also holds different relevant and irrelevant information. Feature extraction procedure provides systematic approach to extract relevant information and simultaneously discard irrelevant one. To choose the relevant characteristics, time frequency decomposition technique is utilized as widely used method. The Discrete Wavelet Transform (DWT) is chosen for MI task categorization based on the different literature which suggests that DWT is suitable technique for non-stationary signal such as EEG. This technique localizes the changes in frequency of signal over time and thus both time and frequency information is taken into consideration unlike CSP features with reduced computational complexity. However, Short Time Fourier Transform is also similar but doesn't provide amplitude fluctuation information and reduced resolution so DWT becomes popular measure [35], [36]. In the current work, the signal is decomposed into 5 level wavelets with mother wavelet as 'dB 4'. As the name suggests that the wavelet has $(N/2 = 4/2)$ 2 vanishing moments with 5 detailed coefficients and one approximate coefficient. The DWT of level 5 is chosen as per the frequency range of interest, 0-100 Hz. See Table I for EEG sub-band frequency range in 5 level DWT decomposition.

From the literature, γ and β waves are responsible for event related synchronization and desynchronization of motor neurons in EEG signal hence, these are taken for further processing in feature extraction of EEG signal. Statistical features of

TABLE I

FREQUENCY RANGE OF DETAILED COEFFICIENTS IN 5 LEVEL DWT

Name of Coefficient	Frequency Range (Hz)	Approximate EEG Frequency band
D1	50-100	Noise
D2	25-50	Gamma
D3	12.5-25	Beta
D4	6.25-12.5	Alpha
D5	0-6.25	Theta + Delta

wavelets are powerful technique of feature extraction. Different statistical features such as standard basic statistics (mean, median, standard deviation etc.) and higher order statistics (entropy, kurtosis, skewness etc.) are explored for EEG signal processing for different applications [14], [37] [38]. It provides better results than only coefficient based features and hence statistical features of wavelet coefficients of chosen sub bands with energy are calculated. The statistical features used in the study include mean, standard deviation, variance, maximum value of detailed coefficient along with power of sub-band. Total 10 features are extracted in the study for each trial. Feature vector for each trail is arranged as:

$$[Mean_{\beta}, std_{\beta}, Var_{\beta}, Max_{\beta}, Pow_{\beta}, \dots, Mean_{\gamma}, std_{\gamma}, Var_{\gamma}, Max_{\gamma}, Pow_{\gamma}]_{(1 \times 10)}$$

These features are further used in feature selection method to get optimal set of features.

3) Feature Selection Using Optimization: The idea of feature selection becomes important to obtain maximum information from minimum features so as to reduce complexity and improve efficiency of classification. Wrappers method of feature selection is most feasible for classifier based feature selection problem. The current work also uses Wrappers method of feature selection. However, Wrapper method is used for feature selection but, the selected features are further utilized in different classifiers to improve MI task classification accuracy [39].

Harmony search (HS) based feature selection technique is used in the current work due to its advantages over other traditional techniques. This is population based approach in which, the new harmony encodes the feature subset based on the selection of recent best feature stored in the harmony memory. Moreover, it can identify multiple best feature subsets than the single one which reduces the chance to get stuck in local minima [40].

The method is inspired by improvisation process of music players. The stochastic nature of the algorithm provides more exploration in the search space and less chances to stuck in the local minima. This property makes the harmony search method more advantageous than the traditional method. Another advantage of the method is that, it is non-sensitive to initial values. Keeping this in mind, harmony search method is utilized for feature selection in the current work. In any optimization method, two main components are objective functions and rules that update the solution. In the current research, the aim is to use optimization to maximize the

TABLE II

SIMULATION PARAMETERS FOR HARMONY SEARCH OPTIMIZATION

Parameter	Value
Number of Features	10
Class	2
Harmony Memory Size (HMS)	100
Harmony Memory Consideration Rate (HMCR)	0.9
PAR	0.4
Number of Iteration	25

classification accuracy. The fitness is calculated as:

$$Fitness = Maximize[Accuracy(F_k)] \quad (2)$$

Here the fitness is function of Features (F_k) which are also known as decision variable. Therefore, selection of features plays an important role. Initially to select the features, binary vectors are generated randomly having the size similar to Feature space. In this method, musicians are decision variables for the fitness function; Notes represent values of the decision variable; Harmonies are number of populations; fitness function is the cost function that needs to be minimized of maximized; and Harmony memory represents the set of good solutions. The algorithm also uses parameters like Harmony memory consideration rate and Pitch Adjustment Rate. These parameters determine the selection or widening the search space and selection of neighbouring values. The Algorithm 1 represents the stepwise evaluation of feature selection method. Input to the algorithm includes feature vector, class labels, simulation parameters, max iterations and count of the features. The algorithm provides the optimal feature subset that will provide maximum classification accuracy as per the objective function. Initially, fitness function is defined and classifier output in the form of accuracy is obtained from Weighted KNN classifier. Based on count of number of features, initial harmony positions are generated. Cost, i.e. fitness value, initialized as empty matrix. In the first for loop, populations are initialized randomly and for each set cost have been calculated. Best cost and best solutions are preserved for evaluation in next iterations. Harmony search optimization is started, for the selected maximum number of iterations, new harmony values are generated and these are compared with HMCR. Since HMCR is higher (near to 1), new solutions will be obtained from existing HM, otherwise it will be generated randomly. The PAR value decides the mutation similar to Genetic Algorithm however HS algorithm uses members of HM for this purpose. After the whole procedure, new selected features are made available and the fitness is evaluated and saved in fitness vector. The whole process is continued until the desired iteration has been achieved and stopping criteria are met. The output of the algorithm provides a set of optimal features with maximum accuracy.

Simulation Parameters: Simulation parameters used for feature selection in harmony search optimization are given in II. Input arguments are listed in which number of features are, extracted features from MI dataset; Class represents categorization problem (Hand/Feet movement); Harmony Memory Size is the limit up to which harmony can store the previous played notes; Harmony Memory Consideration rate is the ran-

domness on choice of feature selection; PAR is the threshold value for updating feature and number of iteration is the max limit of run to update the best solution.

The parameters such as HMS, HMCR and PAR are selected based on the experiment. The various combinations are tested for parameter tuning. The selected parameter results in optimal feature selection that maximizes the accuracy. Then, the change in parameter may also increase the accuracy but at the cost of higher computation or higher complexity. The presented combination provides trade-off between accuracy and complexity. The objective function converges for given set of parameters.

Algorithm for Feature Selection Using Harmony Search

Input: FeatureVector, Class, PAR, HMS, HMCR, MaxIt, Number of features(n) $A = \{a_1, a_2, \dots, a_{10}\}$ of integers

Output: Best feature subset, Best cost

FitnessFunction $\leftarrow$ classification accuracy from Weighted KNN

Harmony Position (Pos) = [];

Cost = [];

for $i \leftarrow 1$ **to** HMS **do**

$Pos = rand(0, 1)_n$

 SelectedFeaturePos $\leftarrow$ (Pos == 1)

 FeatureSet $\leftarrow$ FeatureVector (:,SelectedFeaturePos)

 Cost = FitnessFunction (FeatureSet, Class)

end

HM $\leftarrow$ Sort(Cost)

Best Sol $\leftarrow$ HM(First Value)

for itr $\leftarrow 1$ **to** MaxIt **do**

 NewHarmony $\leftarrow$ empty Harmony of size n

for $k = 1$ **do**

 NewHarmony.Pos $\leftarrow$ random number of size n

for $j \leftarrow 1$ **to** n **do**

if GeneratedRandomNumber < HMCR **then**

$r \leftarrow$ RandomNumber of size 1 to HMS

 NewHarmony(k).Pos(j) $\leftarrow$ HM(r).Pos(j)

else if GeneratedRandomNumber < PAR

then

 NewHarmony(k).Pos(j) $\leftarrow$ 1-HM(k).Pos(j)

end

end

end

 SelectedFeaturePos $\leftarrow$ (NewHarmony(k).Pos(j) == 1)

 FeatureSet $\leftarrow$ FeatureVector (:,SelectedFeaturePos)

 Cost = FitnessFunction (FeatureSet, Class)

end

 HM $\leftarrow$ Merge HM and NewHarmony

 SortOrder $\leftarrow$ Sort(Cost)

 HM $\leftarrow$ HM(SortOrder)

 Truncate HM upto given HMS

 BestSol = HM(Intial Value) BestCost $\leftarrow$ BestSol

end

4) Classification: The classification operation is performed over extracted and selected features. To select the best classifier, well known classification techniques which best suits for

TABLE III
PARAMETERS USED IN TREE CLASSIFIER

Parameter	Bagged Tree
Max number of splits	20
Number of Learners	30
Learner type	Decision Tree
Learning rate	0.1
Subspace dimension	1
Ensemble method	Bag (Random Forest)
Parameter	Complex Tree
Max number of splits	100
Splitting criteria	Gini's Diversity Index[M1]

MI task classification is considered. From the literature, KNN and Tree based classifiers are best among other classifiers hence, these classifiers with their variants have been chosen. The 10 dimensional feature vector is obtained for each subject used for classification of hand and feet moment hence, termed as binary class classification problem. Then 5-fold cross validation is employed for classification.

• Tree Classifier

Tree based classification is non parametric approach heuristic search technique for better classification. However, the method has lower computational complexity and has capability to solve overfitting problem. Irrelevant attribute have less impact on classification accuracy for this classifier. Choice of number of splits make the tree simple to complex. Some ensemble methods also provide enough coverage of features for better classification. Complex tree has maximum splits as 100. As the decision tree is prone to variance and bias so, bagged and subspace tree based methods are used. Bagged trees are ensemble classification tree which select the number of features from split criteria and implement random forest algorithm (generally) on each split, combine the solution to provide classification result [41]. Parameters used in tree classifiers are shown in III. The parameters are the standard parameters presented in the classifiers.

• KNN Classifier

KNN is proximity based lazy classifier which works on instance-based learning method. Instead of building global model best fitted for entire training data, it works on local model and local information-based classification. This might be susceptible to noise but provides flexible decision boundary unlike tree and other rule-based classifiers. Although KNN provides flexible decision boundaries, it very prone to features of interest if not normalized. Features should be in some compatible range to obtain better classification. KNN have many variants based on values of K and distance measurement approach between instances. Majority voting approach is adopted to predict the class of test data:

$$\text{MajorityVotingC} = \operatorname{argmax}_v K(v = C_i) \quad (3)$$

where C_i is the class, v is the label from one of the nearest neighbor, function $K(\cdot)$ returns the binary value of 1 if true else 0 [34]. In the current study, Fine KNN, Weighted KNN, Subspace KNN are adopted for investigation. These KNN variants are chosen due to their optimal K nearest neighbors' value with change in distance measurement technique suitable for the dataset which rejects the outliers efficiently. Table IV shows

TABLE IV
PARAMETERS USED IN KNN CLASSIFIER

Parameter	Fine KNN	Weighted KNN
Number of Neighbours	2	10
Distance Metric	Euclidian	Euclidian
Distance Weight	Equal	Equal
Parameter	Subspace KNN	
Maximum Number of Splits	20	
Number of Learners	30	
Learner Type	Nearest Neighbour	
Learning Rate	0.1	
Subspace Dimension	5	
Ensemble Method	Subspace	

TABLE V

RESULTS OF β AND γ DWT FEATURES FOR MI TASK CLASSIFICATION

Subject Name	Complex Tree	Fine KNN	Weighted KNN	Bagged TREE	Sub-Space KNN
S01	85.9	92.9	93.6	92.5	92.2
S02	87.9	92.1	93.3	93.1	91.5
S03	88.9	92	91.9	91.3	91.6
S04	87.7	90.9	92.3	92.7	89.9
S05	86.3	92.1	91.9	91.8	92.5
S06	87.4	91.3	92.3	92	90.1
S07	86.9	92.2	91	91.5	91.3
S08	87.3	92.7	92.8	93.1	92.6
S09	88.6	92.6	92.9	93.2	94
S10	89	94	92.9	92.7	92.9
AVG	87.59	92.28	92.49	92.39	91.86
AUC	0.92	0.93	0.98	0.98	0.97

Parameters used in the KNN classification. Which are the standard parameters presented in the classifiers. The results are shown in the subsequent sections.

V. RESULTS AND DISCUSSION

The experiment was performed on MATLAB software with system configuration intel core i5 processor and 8 GB RAM. Open source dataset is tested for offline analysis of the proposed scheme of motor imagery task classification.

The experiment was performed on motor imagery task classification dataset as presented in section IV-A. To achieve the promises set in section III, different experiments have been performed. For improved classification accuracy, statistical wavelets of beta and gamma frequency band are extracted and analyzed, the obtained results are presented further in this section. The later part of this section represents the result obtained from harmony search optimization based feature selection method. Comparative analysis of proposed approach with existing approaches also presented in this section. The results used for the comparison in VII, VIII are directly taken from the research papers. The significance of proposed method also analyzed with statistical test presented in V-D.

A. Performance of Beta- Gamma sub-Band Based Features

This experiment was performed on MI dataset discussed in section IV-A, beta and gamma frequency based statistical DWT (SDTW) technique of feature extraction is employed. The experiment have been performed on all 10 available subjects in which left hand and feet movement have to be classified. The resultant feature matrix is obtained after SDWT

TABLE VI
PRECISION, RECALL AND F-SCORE USING WEIGHTED KNN CLASSIFIER

Subject	Precision	Recall	F-Score
S01	0.9	0.95	0.92
S02	0.929	0.912	0.923
S03	0.924	0.965	0.94
S04	0.944	0.929	0.936
S05	0.924	0.926	0.923
S06	0.919	0.912	0.917
S07	0.932	0.9	0.918
S08	0.907	0.954	0.924
S09	0.947	0.953	0.951
S10	0.925	0.928	0.93

feature with dimension 1260×10 for each subject. The experiment was performed for 5- fold cross validation of each subject.

The results shown in V represents comparative analysis of variants of Tree and KNN classifier. From the Table V weighted KNN shows better performance among all the classifiers for approximately all the subjects. The better performance of Weighted KNN is further utilized in feature selection method.

The precision recall and F Score are also reported for Weighted KNN classifier. The results presented in Table VI are evaluated with this classifier because it shows lesser complexity, higher performance and doesn't introduce bias in the algorithm. Which is further utilized in the paper.

From the results, Area Under the Curve (AUC) is shown in Table V, AUC for weighted KNN and bagged tree both are 0.98, which shows the reliability for further use.

B. Performance of Harmony Search Based Feature Selection Method

The another objective of the paper was to evaluate the performance of optimization technique for feature selection in MI task classification. Harmony search optimization is used in the current work for feature selection due to the advantages presented in IV-B.3. The experiment was performed 25 times for each subject and average accuracy has reported. In the experiment, feature selection is applied to whole dataset as we are using 5 fold cross validation technique. This ensure there would be no bias in the experiment due to lack of representation of samples in training and testing set. The average accuracy from all the 5 fold data evaluation has reported. Fig. 2 shows the average accuracy obtained after 25 runs. The Fig. 2 also compare accuracy before and after optimization for Weighted KNN classifier. It can be said that feature selection using harmony search method shows significant impact on classification accuracy. Feature selection also reduces the feature space upto 6 features from 10 features.

Frequency of features selected in 25 runs are shows that the higher the occurrence the more the importance, but it can be seen that, no single feature is dominating. The combination of features shows more impact on classification task.

C. Comparative Analysis

This section deals with comparative analysis of proposed algorithm with existing algorithms. Common spatial patterns are widely used features for MI task classification however,

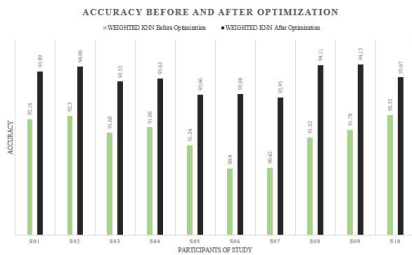


Fig. 2. Impact of feature selection in MI task classification.

TABLE VII
PERFORMANCE COMPARISON OF PROPOSED APPROACH
WITH CSP BASED METHOD

SUBJECT NAME	CSP+RF	CSP+LDA	Proposed
S01	74.56	69.69	93.89
S02	80.52	79.22	94.06
S03	99.3	99.22	93.52
S04	81.85	79.09	93.63
S05	65.44	63.1	93.06
S06	88.46	84.35	93.08
S07	88.33	88.61	92.95
S08	83.28	78.93	94.11
S09	76.64	73.1	94.13
S10	54.61	56.22	93.67

the results for given dataset using CSP method have a lot of scope of improvement. The results of proposed approach are compared with CSP method as presented in table VII. It can be seen that the proposed approach shows higher performance than the CSP based method. In table VIII Wavelet based features are compared with the proposed algorithm. The work reported in comparison used the wavelet coefficients as feature and perform classification and another literature reported results using PCA based feature compression however we are using statistics of wavelets which allows to represent the series of features as a single value.

There is large difference in the results of proposed algorithm and compared algorithms due to the fact that, for the table VII, extracted features are CSP features, which suffer from the drawback of overfitting and also introduce the bias. This method focuses on spatial property of signal not the spectral one however, the proposed approach uses the spectral property of feature, specially the beta and gamma frequency based features that are more related to the MI events. The literature also suggests that event related synchronization and desynchronization happens in the such situations. The statistics of those features provide more clear information than only wavelet based feature or PCA as compared in table VIII. The HS optimization technique helps to get the better combination of feature set out of those features which further improve the performance and reduce the feature space. There are a number of techniques for classification of motor imagery tasks but the proposed algorithm works with less number of features and it also opens the another dimension of research in MI task classification using optimization technique.

Apart from comparison, statistical analysis has been performed to analyze the significance of proposed algorithm. The statistical analysis presented in further subsection.

TABLE VIII
PERFORMANCE COMPARISON OF PROPOSED APPROACH
WITH DWT BASED FEATURES

Subjects	Wavelet Feature [42]	PCA Feature Compression [43]	Feature Selection [44]	Proposed SDWT Features	Proposed Feature selection
S01	45.3	59.3	59.3	92.16	93.89
S02	71.8	60.9	56.2	92.3	94.06
S03	56.2	61.3	57.5	91.68	93.52
S04	46.8	—	60	91.88	93.63

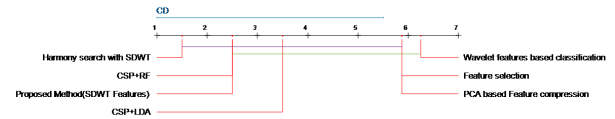


Fig. 3. Results of statistical test.

D. Statistical Analysis

Statistical analysis is performed for the different algorithms used for classification of motor imagery task. The Friedman test is employed for statistical comparison on proposed method with other existing methods because this test allows to compare multiple algorithms [45]. Null hypothesis suggests that the methods are same with alternative hypothesis is that the methods are significantly different. Nemenyi post hoc analysis is employed to test the best method among available ones [46].

The results from Friedman test shows the “p” value 0.0029 which is less than significance level 0.05. since $p < 0.05$, null hypothesis can be rejected and can be concluded that methods are statistically different. Further, Nemenyi post hoc analysis results are presented in Fig. 3. The CD is the confidence value obtained from Nemenyi post hoc analysis. The Fig. 3 represents the ranking of methods. The best method has rank 1, next best gets rank 2 and so on. From the figure it can be seen that the proposed feature selection approach gets 1st rank among all methods. The proposed approach with β and γ features also gets 2nd rank which shows the significance of the proposed method for further use.

VI. CONCLUSION

This article studies motor imagery task classification with beta and gamma frequency based sub-bands. These features are found to be very efficient for classification of motor imagery task. Initially 0.5 - 100Hz band pass filter is used to make the signal compatible with frequency bands of brain signal with 5 level DWT of dB4 wavelet. The Wavelets coefficients of choice of frequency bands are extracted and further statistics of these are calculated. These extracted features outperform than other features. Further feature selection has been employed to find the relevant features. This method improves the classification accuracy using weighted KNN. This shows that HS based optimal feature selection method improves the overall performance of motor imagery task classification. The DWT based features under the influence of gamma frequency have measurable impact in motor imagery classification task. The proposed method in this article for feature extraction and selection can be improved further with parameter tuning for better classification accuracy of MI task. Gamma frequency

for MI task will also be further explore to find the dominance of specific frequency band.

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